

Question 1 Solution

Sample mean:

$$\bar{y} = \frac{24.8 + 25.0 + 24.6 + 24.4}{4} = \frac{98.8}{4} = 24.70$$

Question 1(a)

$$x_1 = \bar{y} - T_{\text{target}} = 24.70 - 24 = \mathbf{0.70}$$

$$\begin{aligned} x_2 &= \frac{1}{N} \sum_{i=1}^N (y[i] - \bar{y})^2 \\ &= \frac{1}{4} [(24.8 - 24.7)^2 + (25.0 - 24.7)^2 + (24.6 - 24.7)^2 + (24.4 - 24.7)^2] \\ &= \frac{1}{4} [0.01 + 0.09 + 0.01 + 0.09] \\ &= \mathbf{0.05} \end{aligned}$$

$$x_3 = y[4] - T_{\text{target}} = 24.4 - 24 = \mathbf{0.40}$$

Question 1(b)

$$\begin{aligned} s &= w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3 + b \\ &= 1.2 \times 0.70 + 0.6 \times 0.05 + (-1.0) \times 0.40 + (-0.3) \\ &= 0.84 + 0.03 - 0.40 - 0.30 \\ &= \mathbf{0.17} \end{aligned}$$

Question 1(c)

$$p = \frac{1}{1 + e^{-s}} = \frac{1}{1 + e^{-0.17}} = \frac{1}{1 + 0.8437} = \frac{1}{1.8437} \approx \mathbf{0.54}$$

Since $p = 0.54 \geq 0.5$, the decision is **AC ON**.

Question 2 Solution

Question 2(a)

$$\text{pred} = \text{input} \times \text{weight} = 9 \times 0.10 = \mathbf{0.900}$$

Question 2(b)

$$\delta = \text{pred} - \text{goal} = 0.900 - 1.0 = \mathbf{-0.100}$$

$$\text{loss} = \delta^2 = (-0.100)^2 = \mathbf{0.010}$$

Question 2(c)

$$\Delta w = \text{input} \times \delta = 9 \times (-0.100) = \mathbf{-0.900}$$

$$\begin{aligned} \text{new_weight} &= \text{weight} - \alpha \times \Delta w \\ &= 0.10 - 0.10 \times (-0.900) \\ &= 0.10 + 0.090 \\ &= \mathbf{0.190} \end{aligned}$$

Question 3

Question 3(a)

$$\mathbf{q}_{x_1} = \text{round}(x_1/s_x) + z_x = \text{round}(0.48/0.2) - 1 = 2 - 1 = 1$$

$$\mathbf{q}_{x_2} = \text{round}(x_2/s_x) + z_x = \text{round}(-0.22/0.2) - 1 = -1 - 1 = -2$$

$$\mathbf{q}_{w_1} = \text{round}(w_1/s_w) + z_w = \text{round}(0.18/0.06) + 0 = 3$$

$$\mathbf{q}_{w_2} = \text{round}(w_2/s_w) + z_w = \text{round}(-0.09/0.06) + 0 = -2$$

Question 3(b)

From Question 3(a), we have

$$\mathbf{q}_x = \begin{pmatrix} \mathbf{q}_{x_1} \\ \mathbf{q}_{x_2} \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \quad \mathbf{q}_w = \begin{pmatrix} \mathbf{q}_{w_1} \\ \mathbf{q}_{w_2} \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

Then,

$$\begin{aligned} \mathbf{acc} &= (\mathbf{q}_x - z_x) \cdot \mathbf{q}_w \\ &= \left(\begin{pmatrix} 1 \\ -2 \end{pmatrix} - (-1) \right) \cdot \begin{pmatrix} 3 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \end{pmatrix} \\ &= 2 \times 3 + (-1) \times (-2) \\ &= 6 + 2 \\ &= \mathbf{8} \end{aligned}$$

Question 3(c)

$$\begin{aligned} q_y &= \text{round}(\mathbf{acc} \cdot M) + z_y \\ &= \text{round} \left(8 \cdot \left(\frac{s_x \cdot s_w}{s_y} \right) \right) + 0 \\ &= \text{round} \left(8 \cdot \left(\frac{0.2 \cdot 0.06}{0.03} \right) \right) \\ &= \text{round}(8 \cdot (0.4)) \\ &= \mathbf{3} \end{aligned}$$

Finally,

$$\begin{aligned} \hat{r} &= q_y \cdot s_y \\ &= 3 \cdot 0.03 \\ &= \mathbf{0.09} \end{aligned}$$

Question 4

Question 4(a)

$$\text{Efficiency}_A = \frac{\Delta\text{Cost}_A}{I_A} = \frac{1200 + 800}{40} = \frac{2000}{40} = \mathbf{50}$$

$$\text{Efficiency}_B = \frac{\Delta\text{Cost}_B}{I_B} = \frac{900 + 600}{20} = \frac{1500}{20} = \mathbf{75}$$

$$\text{Efficiency}_C = \frac{\Delta\text{Cost}_C}{I_C} = \frac{1500 + 300}{60} = \frac{1800}{60} = \mathbf{30}$$

Question 4(b)

According to greedy pruning rule, because $\text{Efficiency}_B(75) > \text{Efficiency}_A(50) > \text{Efficiency}_C(30)$, we should prune **Layer B** in this iteration. The reason is near definitional from a cost-driven pruning perspective: Layer B offers the **highest computational savings per unit of importance loss**, thus it offers the best deal for our objective.